

Comparative Performance Analysis of Lossless MRI Image Compression Algorithms for Efficient Medical Image Processing

Parvathi A (U2301164)

Pavithra N Pradeep (U2301167)

Sivadath Siju (U2301200)

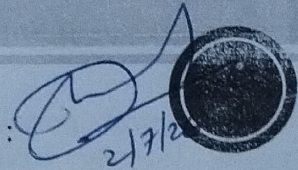
Sourav V P (U2301204)

Guided by :

Mr. Dhanesh M S

Asst. Professor

Dept. of ECE



INTRODUCTION

- Medical imaging technologies such as MRI generate a large volume of high-resolution diagnostic images every day.
- Hospitals require efficient storage and rapid transmission of these images through PACS and telemedicine networks.
- Since MRI images are used for diagnosis, every pixel contains valuable clinical information, making lossless compression essential.
- FPGA-based hardware acceleration provides faster compression while maintaining diagnostic image quality.

Comparative Performance Analysis of Lossless MRI Image Compression Algorithms for Efficient Medical Image Processing

Parvathi A (U2301164)

Pavithra N Pradeep (U2301167)

Sivadath Siju (U2301200)

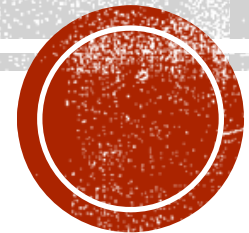
Sourav V P (U2301204)

Guided by :

Mr. Dhanesh M S

Asst.Professor

Dept. of ECE



INTRODUCTION

- Medical imaging technologies such as MRI generate a large volume of high-resolution diagnostic images every day.
- Hospitals require efficient storage and rapid transmission of these images through PACS and telemedicine networks.
- Since MRI images are used for diagnosis, every pixel contains valuable clinical information, making lossless compression essential.
- FPGA-based hardware acceleration provides faster compression while maintaining diagnostic image quality.

OBJECTIVE

- To develop an efficient lossless compression system for MRI images by comparing performance of two state-of-the-art compression algorithms and implementing the most suitable algorithm on an FPGA platform for high-speed, resource-efficient medical image processing.
- Evaluate the algorithms based on Compression Ratio (CR), Bits Per Pixel (BPP), image quality, computational complexity, and execution time.
- Implement the best-performing algorithm considering both software performance and hardware feasibility.
- Achieve reduced storage requirements, faster image transmission, lower hardware resource utilization, and improved throughput for real-time medical imaging applications.

PROBLEM STATEMENT

Despite advances in medical image compression, several challenges remain:

- Increasing use of MRI imaging generates large volumes of high-resolution data, creating storage and transmission challenges.
- PACS (Picture Archiving and Communication System) requires efficient storage and fast retrieval of medical images while maintaining diagnostic integrity.
- Existing lossless compression methods require high computational and memory resources.
- Preserving diagnostic information while achieving efficient compression is critical.
- There is a need for an FPGA-based compression framework that balances compression efficiency with hardware performance.

Export Data

Please note when exporting: patient-related information must be protected

Patient	Series description	Instances
☒	t1_se_sag	29
☒	t2_space_dark-fluid_s...	160
☒	FUJI Basic Text SR for ...	1
☒	ep2d_diff_3scan_trac...	70
☒	ep2d_diff_3scan_trac...	35
☒	t2_tse_dark-fluid_tra	35
☒	pd+t2_tse_tra	68
☒	t2_swi_tra_p2_2mm...	80
☒	t2_swi_tra_p2_2mm...	80
☒	t2_swi_tra_p2_2mm...	80
☒	t2_swi_tra_p2_2mm...	73
☒	AAhead_scout_MPR...	5
☒	t1_mprage_tra_p2_iso	192
☒	t2_tse_cor	36
☒	+C t1_mprage_tra_p...	192
☒	+C t1_mprage_tra_p...	192
☒	+C t1 mprage tra p...	232

514 MB, 1810 Instances

Clear List

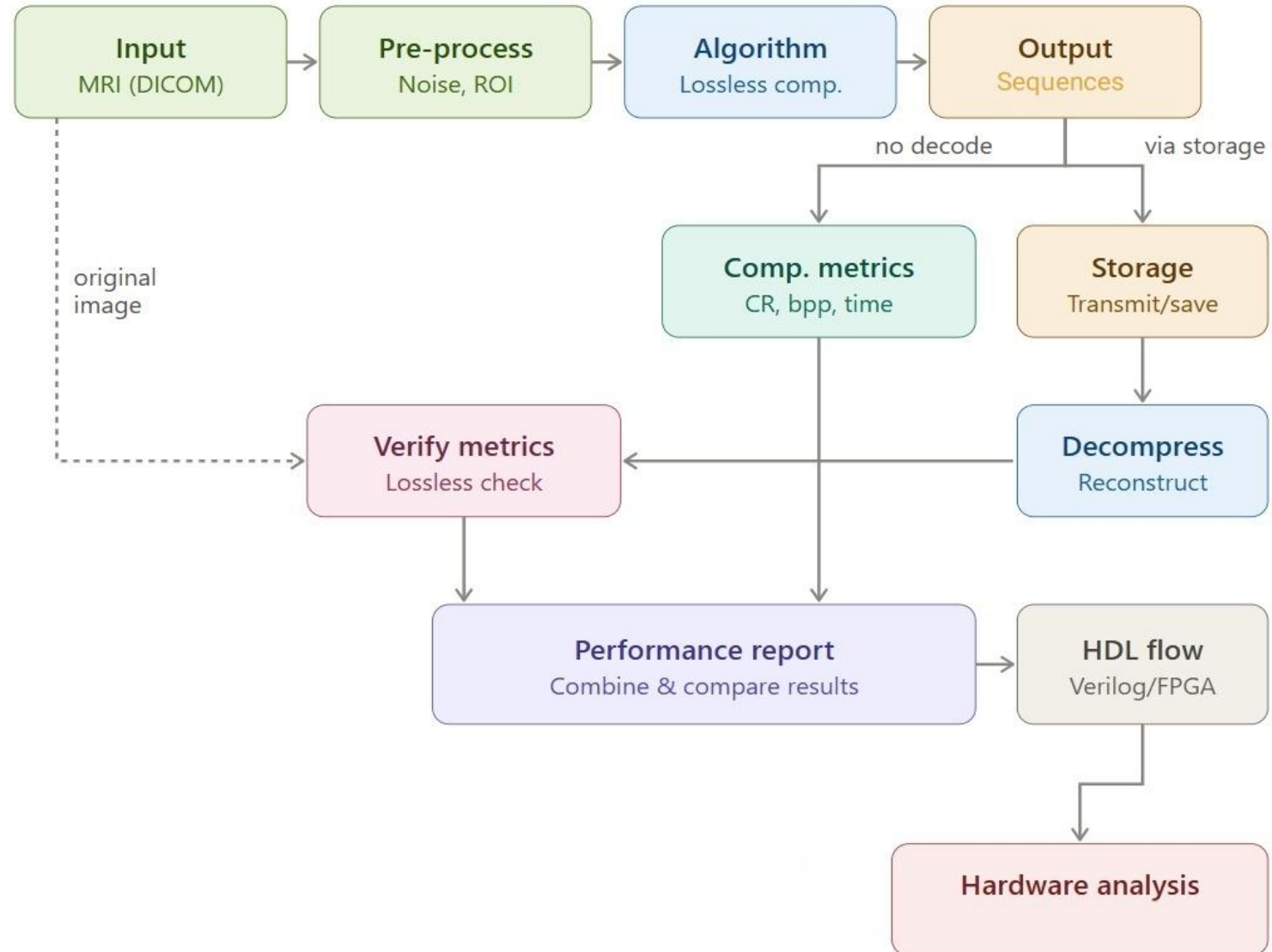
Export

- A single patient examination can contain multiple imaging sequences such as T1, T2, DWI, SWI, and MPRAGE,
- Each sequence comprising approximately 20–200 image instances.
- Single MRI instances can be as large as 4 MB,
- One real patient study contained 1,810 image instances occupying 514 MB.
- In comprehensive or high-resolution studies, the total MRI data size can even reach several gigabytes.
- This large and growing volume of medical imaging data creates significant challenges in **storage, transmission, and archival**, thereby necessitating efficient lossless compression techniques that preserve diagnostic information.

PROPOSED SOLUTION

- Determine the optimal lossless medical image compression algorithm by evaluating the compression efficiency and computational performance of JPEG-LS and HENCE.
- Identification can be done by performing the comparison based on Compression Ratio (CR), Bits Per Pixel (BPP), Compression Time, Decompression Time, and Lossless Reconstruction Accuracy.
- Implement the selected algorithm on an FPGA platform using Vitis HLS.
- Evaluate the FPGA implementation based on Throughput, Latency, Resource Utilization (LUTs, DSPs, BRAM), and Power Consumption.

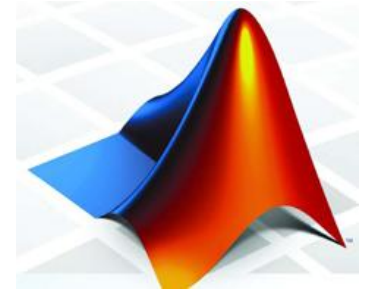
BLOCK DIAGRAM



HARDWARE AND SOFTWARE

Software Tools:

- MATLAB
- Vitis HLS



Hardware Components:

- Kria KV260

Provides a Zynq UltraScale+ MPSoC with 256K logic cells, 4 GB DDR4 memory, and high-speed MIPI, HDMI, USB, and Gigabit Ethernet connectivity.



WORK DIVISION

- **Sourav V P- Baseline Algorithm Implementation:** Develop and implement the baseline lossless compression algorithm in MATLAB.
- **Parvathi A- HENCE Algorithm Implementation:** Study the HENCE framework and implement the HENCE-based compression algorithm in MATLAB.
- **Pavithra N Pradeep- Dataset & Performance Evaluation:** Prepare MRI datasets and compare both algorithms using compression ratio, execution time, and other performance metrics.
- **Sivadath Siju- Documentation & Vitis HLS Exploration:** Document project progress and explore Vitis HLS for future C++ implementation and FPGA deployment.

SEMINAR TOPICS

1.Context modeling for lossless 3D/volumetric medical image compression

Explores bilateral context modeling to exploit both intra-slice and inter-slice redundancies for efficient lossless 3D medical image compression.

2.Hardware-efficient Sigmoid/nonlinear function approximation

Investigates hardware-efficient Sigmoid approximation techniques to enable accurate and resource-efficient FPGA implementation of neural compression algorithms.

3.Hardware acceleration of end-to-end learned image compression

Presents an FPGA-accelerated end-to-end learned image compression framework using algorithm–architecture co-optimization for high throughput and efficient resource utilization.

4.Efficient entropy modeling for learned lossless compression

Introduces a multirate progressive entropy model to improve probability estimation and compression efficiency in learned image compression.

CONCLUSION

- Lossless compression is essential for preserving the diagnostic quality of MRI images while reducing storage and transmission requirements.
- A comparative performance analysis of two lossless compression algorithms will be conducted to identify the most efficient approach.
- The selected algorithm will be implemented on an FPGA to achieve high-speed processing, low hardware resource utilization, and improved throughput.
- The proposed work aims to provide an efficient and reliable solution for real-time medical image compression in healthcare systems.

REFERENCE

- [1] J. Chen, Q. Chen, H. Zhang, W. Chen, W. Luo and F. Yu, "HENCE: Hardware End-to-End Neural Conditional Entropy Encoder for Lossless 3D Medical Image Compression," in IEEE Access, vol. 12, pp. 186520-186534, 2024, doi: 10.1109/ACCESS.2024.3514215.
- [2] F. Liu, X. Chen, Z. Liao and C. Yang, "Adaptive Pipeline Hardware Architecture Design and Implementation for Image Lossless Compression/Decompression Based on JPEG-LS," in IEEE Access, vol. 12, pp. 5393-5403, 2024, doi: 10.1109/ACCESS.2024.3350344.
- [3] X. Liu, M. Wang, S. Wang and S. Kwong, "Bilateral Context Modeling for Residual Coding in Lossless 3D Medical Image Compression," in IEEE Transactions on Image Processing, vol. 33, pp. 2502-2513, 2024, doi: 10.1109/TIP.2024.3378910.
- [4] V. de A. Bosso, R. M. Nardini, M. A. de A. de Sousa, S. D. dos Santos, and R. Pires, "An Area-Efficient and Low-Error FPGA-Based Sigmoid Function Approximation," *Applied Sciences*, vol. 15, no. 21, p. 11551, Oct. 2025, doi: <https://doi.org/10.3390/app152111551>.
- [5] C. Li, S. Yin, C. Jia, F. Meng, Y. Tian and Y. Liang, "Multirate Progressive Entropy Model for Learned Image Compression," in IEEE Transactions on Circuits and Systems for Video Technology, vol. 34, no. 8, pp. 7725-7741, Aug. 2024, doi: 10.1109/TCSVT.2024.3376704.